DP014
Physics 1
Semester I
Session 2023/2024
2 hours

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TRIAL 1

PHYSICS UNIT

DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

INSTRUCTIONS TO CANDIDATE:

The question paper consists of 8 questions.

Answer **all** questions.

The use of electronic calculator is permitted.

QUESTION	MARKS
1	/8
2	/13
3	/9
4	/12
5	/9
6	/12
7	/17
TOTAL	/80

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$
4. $s = \frac{1}{2}(u + v)t$
5. $v = u - gt$
6. $v^2 = u^2 - 2gs$
7. $s = ut - \frac{1}{2}gt^2$
8. $p = mv$
9. $J = F\Delta t$
10. $J = \Delta p = mv - mu$
11. $f_s \leq \mu_s N$
12. $f_k = \mu_k N$
13. $W = \vec{F} \cdot \vec{s} = Fs\cos\theta$
14. $K = \frac{1}{2}mv^2$
15. $U = mgh$
16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$
17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$
18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$
19. $s = r\theta$
20. $v = r\omega$
21. $a_t = r\alpha$
22. $\omega = \omega_\circ + \alpha t$
23. $\theta = \omega_\circ t + \frac{1}{2}\alpha t^2$
24. $\omega^2 = {\omega_\circ}^2 + 2\alpha\theta$
25. $\theta = \frac{1}{2}(\omega_\circ + \omega)t$
26. $\tau = rF\sin\theta$
27. $\frac{Q}{t} = -kA\left(\frac{\Delta \mathrm{T}}{L}\right)$
28. $v_{rms} = \sqrt{< v^2 >}$
29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$
30. $\Delta U = Q - W$

Electron volt *Elektron volt*	1 eV	$= 1.6 \times 10^{-19}$ J
Constant of proportionality for Coulomb's law *Pemalar hukum Coulomb*	$k = \frac{1}{4\pi\varepsilon_\circ}$	$= 9.0 \times 10^{9}$ N m^{2} C^{-2}
Atmospheric pressure *Tekanan atmosfera*	1 atm	$= 1.013 \times 10^{5}$ Pa
Density of water *Ketumpatan air*	ρ_w	$= 1000$ kg m^{-3}

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum *Laju cahaya dalam vakum*	c	$= 3.00 \times 10^{8}$ m s^{-1}
Permeability of free space *Ketelapan ruang bebas*	μ_0	$= 4\pi \times 10^{-7}$ H m^{-1}
Permittivity of free space *Ketelusan ruang bebas*	ε_0	$= 8.85 \times 10^{-12}$ F m^{-1}
Electron charge magnitude *Magnitud cas elektron*	e	$= 1.60 \times 10^{-19}$ C
Planck constant *Pemalar Planck*	h	$= 6.63 \times 10^{-34}$ J s
Electron mass *Jisim elektron*	m_e	$= 9.11 \times 10^{-31}$ kg $= 5.49 \times 10^{-4}$ u
Neutron mass *Jisim neutron*	m_n	$= 1.674 \times 10^{-27}$ kg $= 1.008665$ u
Proton mass *Jisim proton*	m_p	$= 1.672 \times 10^{-27}$ kg $= 1.007277$ u
Deuteron mass *Jisim deuteron*	m_d	$= 3.34 \times 10^{-27}$ kg $= 2.014102$ u
Molar gas constant *Pemalar gas molar*	R	$= 8.31$ J K^{-1} mol^{-1}
Avogadro constant *Pemalar Avogadro*	N_A	$= 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant *Pemalar Boltzmann*	k	$= 1.38 \times 10^{-23}$ J K^{-1}
Free-fall acceleration *Pecutan jatuh bebas*	g	$= 9.81$ m s^{-2}
Atomic mass unit *Unit jisim atom*	1 u	$= 1.66 \times 10^{-27}$ kg $= 931.5 \frac{MeV}{c^2}$

1. a) To overtake a lorry, a car speeds up from 60 km h^{-1} to 120 km h^{-1} in 5 seconds. Calculate the acceleration of the car in S.I. units.

[3 marks]

b) Three vectors are given as A = 15 m, B = 20 m and C = 30 m as shown in **Figure 1**.

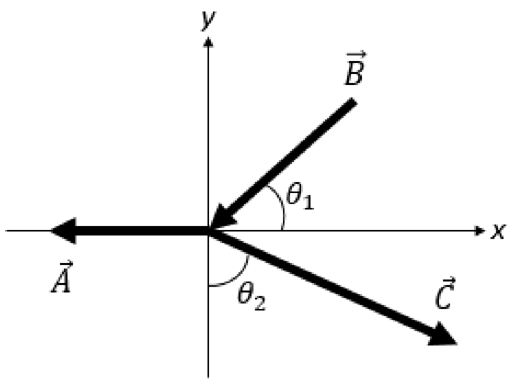

Figure 1

If $\theta_1 = 45°$ and $\theta_2 = 60°$, calculate the resultant vector.

[5 marks]

2. a) An object is thrown vertically upwards from a point on the ground with speed of *u* until it reach back to the ground. Sketch the graph for the motion with the situation below.

i) displacement vs time.

ii) velocity vs time.

iii) acceleration vs time.

[6 marks]

b) A small pebble is thrown upward from a cliff with an initial velocity 20 m s^{-1}. Calculate

i) Maximum height reached from its initial position.

ii) Time taken to reach the maximum height.

iii) Time taken to reach a point 25 m below the initial point.

[7 marks]

3. a) A 8.0 kg toy car moving to the right at 3.35 m s^{-1} strikes a stationary ball with mass of 5.0 kg. After the collision, the toy car rebounds with a speed of 5.45 m s^{-1} with an angle of 30^0 above the horizontal. Determine the final velocity of the ball.

[7 marks]

b) A ball with a mass of 100 g strikes a wall perpendicularly at a velocity of 12.0 ms^{-1} and rebounds at a velocity of 8.0 m s^{-1}. Calculate the impulse of the ball on the wall.

[2 marks]

4. Two blocks of mass $m_1 = 300$ g and $m_2 = 500$ g are pushed by a force, F as in **Figure 2**. The coefficient of friction between each block and table is 0.4.

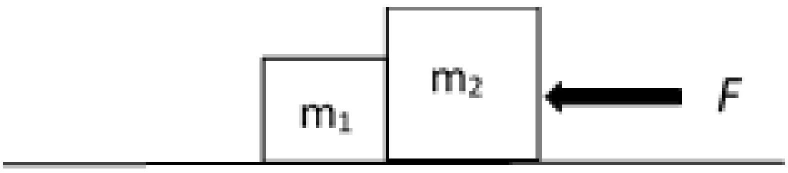

Figure 2

a) Sketch free body diagram for both blocks. Label all the forces.
b) Determine the frictional force due to table for both blocks.
c) Determine the value of F if the blocks are accelerated of $200\ \mathrm{cms}^{-2}$?
d) Determine the forces exerted on m_2 by m_1 ?

[12 marks]

5. a) A car travels at a constant speed around a circular track whose radius is 2.6 km. The car goes once around the track in 360 s. What is the magnitude of the centripetal acceleration of the car?

[3 marks]

b) A string fixed to a ceiling has its end fastened to a ball of mass 200 g. The ball whirls at constant speed in a horizontal circle as a conical pendulum. If the string is of length 32 cm, and makes and angle 30º with the vertical, what is the speed of the ball?

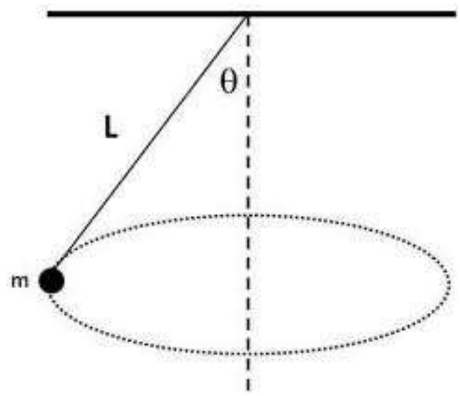

Figure 3

[6 marks]

6. a) A disk 8.00 cm in radius rotates at a constant rate of 1200 rev min^{-1} about its central axis. Determine

i) its angular speed in rad s^{-1}.

ii) the tangential speed at a point 3.00 cm from its centre.

iii) the radial acceleration of a point on the rim.

[6 marks]

b) **Figure 4** shows the forces, F_1 =30 N, F_2 = 10 N and F_3 = 20 N are applied to a rectangle with side lengths is 10 m and width is 6 m. Calculate the resultant (magnitude and direction) torque of all forces about point O.

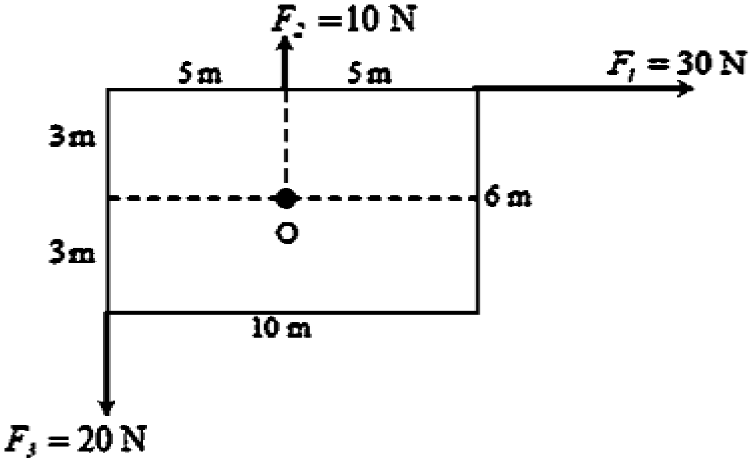

Figure 4

[6 marks]

7. a) One face of an aluminium cube of edge 4 meters is at 60 ºC and the other end is at 0 ºC. All other surfaces are covered by good insulator adiabatic walls.

i) Find the temperature gradient of the aluminium cube

ii) Calculate the rate of heat transfer

iii) Find the amount of heat flowing through the cube in 2 seconds.

(Thermal conductivity of aluminium is 209 W m^{-1} $°C^{-1}$).

[6 marks]

b) A fresh air is composed of nitrogen N_2 and oxygen O_2 is at 40°C.

i) Calculate the temperature, T in SI unit.

ii) Find the rms speed of O_2.

(Given molar mass, M of Oxygen gas = 32 g/mol)

[4 marks]

c)

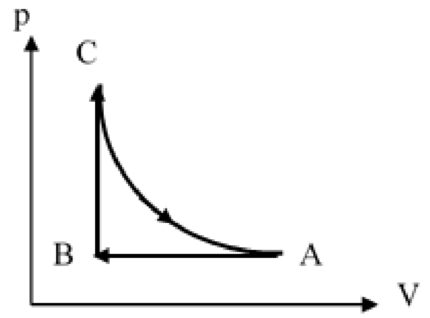

Figure 5

Figure 5 shows p-V graph of an ideal gas undergoes thermodynamics process. Process AB has change in internal energy 300 J from higher temperature to the lower temperature. The work done on the gas in process AB is 135 J and 160 J heat is added to the gas for the process BC. Then, the gas expands isothermally and does 450 J of work in the process CA. Determine

i) the heat in the process AB and state whether the heat is absorbed or released.

ii) the work done and change in internal energy in the process BC.

iii) change in internal energy **and** heat in the process CA.

[7 marks]